

Michele Ferretti

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CURRENT POSITION

Senior Data Engineer, Geospatial
NVIDIA, Munich, Germany

Sept. 2022 - Now

At NVIDIA Autonomous Driving I work on the search and curation infrastructure, serving data at scale to build the next generation of driving applications. In my role, I generate value and impact by fostering collaboration across teams, mentoring peers, and delivering practical solutions that drive business outcomes.

EXPERIENCE

Machine Learning Scientist
Booking.com, Amsterdam, The Netherlands

May 2019 - Sept. 2022

As Machine Learning Scientist in the Geo Product team, I worked on core mapping and location technologies. My responsibilities included:

- developing ML products to exploit spatially-explicit features
- scaling and enriching the spatial data catalogue via automated map-making
- designing, building and maintaining data pipelines and tooling

Visiting Data Scientist
FBK - Fondazione Bruno Kessler, Trento, Italy

Jan. 2018 - June 2018

I researched a novel ML method combining Semantic data with Location-Based social media. I contributed to the development of Python package to facilitate its integration in Data Science applications.

Research Fellow
Mapbox, Remote

May 2017 - Sept. 2017

Embedded with the Cities team, I worked on the design and implementation of 'Smart City' products, with a focus on smart mobility. I regularly attended clients meetings and worked closely with the Cities team leader to achieve valuable outcomes for our clients.
Supervisor: Christina Franken

Research Scientist Intern
IBM Research, Dublin, Ireland

June 2016 - Sept. 2016

With the IBM Smarter Cities lab researchers, I worked on a UNHRF-commissioned Machine Learning application. The internship results have been published in the PKDD paper (Ahmed, 2016).
Supervisors: Dr. Francesco Calabrese and Dr. Rahul Nahir

EDUCATION

Ph.D. in Geography [Fully funded via scholarship]
King's College London, United Kingdom,
Informatics Department - Centre for Urban Science and Progress (CUSP) London,
Geography Department - Geocomputation research domain

Nov. 2015 - May. 2021

I researched the development of 'Smart Cities' technologies, with a focus on geospatial applications. I carried out two industrial internships at the IBM Europe Research lab and

at Mapbox.
Advisor: Dr. Jonathan Reades

Master of Science in Urban Planning 2011 - 2013

Politecnico di Milano, Milano, Italy

Thesis: ‘The Death of Distance in the Age of Urban Sensing’. Final grade: *110/110 cum Laude*.

Advisors: Prof. P. McCann (Rijksuniversiteit Groningen), Prof. V. Fedeli & Prof. I. Mariotti (Politecnico di Milano).

Postgraduate Course on ‘Sistemi Informativi Territoriali e Governo del Territorio’ [Fully funded via scholarship] 2012

Politecnico di Milano, Milano, Italy

Post-graduate courses on ‘GIS and Spatial Analysis for territorial governance’.

Bachelor of Science in Environmental Science 2009-2012

Università degli Studi di Milano-Bicocca, Milano, Italy

Thesis: ‘Landuse classification through Vis-NIR Airborne imagery’. Advisor: Prof. Roberto Colombo.

PROJECTS

King’s Humanitarian Mappers [Awarded Project] Jan. 2016 - 2018

Co-founder & Co-organizer

Drawing from a larger global movement of *digital humanitarians* together with KCL colleagues I run a bi-weekly *mapping marathons* to help researchers and NGOs in preparedness and response initiatives.

ACHILLES [Awarded Project] Nov. 2017

Scientist

ACHILLES, awarded at the CIKM’17 - DataSpark Mobility Challenge, is a web-based tool for transportation planning based on a Multi-Layer Network representation. It allows a non-user expert to rapidly visualise and assess a transport system’s resilience to traumatic events such as floods and congestion.

KCL Code Camp June 2016 - August 2016

Researcher & Developer

Created a comprehensive set of online notebooks to introduce first-year students in the KCL Geocomputation Pathway to Python programming. The course is open source (MIT licensed) and accessible via the [KCL Code Camp](#).

Invisible Cities Nov. 2016

Developer

I helped with the ideation and development of a generative AI project, training a Pix2Pix-based GANN to translate hand-drawn sketches into satellite images of Venice and Los Angeles.

Maptime! Milan Meetup 2014 - 2016

Founder & Organizer

I founded and managed the Geo-focused Meetup chapter, organising workshops and delivering public talks.

LANGUAGES

- Italian Mother Tongue
- English Professional proficiency
- Spanish Limited working proficiency

PUBLICATIONS

- ‘I Know What You Did Next Summer: Challenges in Travel Destinations Recommendation.’ *D. Goldenberg, S. Mizrahi, A. Horowitz, I. Kangas, O. Levkovich, A. Mozzato, M. Schwoerer, M. Ferretti, P. Korvesis, L. Bernardi - RecTour@ RecSys, 2021*
- ‘Weak nodes detection in urban transport systems: Planning for resilience in Singapore.’ *M.Ferretti, L.Pappalardo, G.Barlacchi and L.Lepri - IEEE 5th International Conference on Data Science and Advanced Analytics (DSAA), 2018*
- ‘Ordinary Chinese smart cities: the case of Wuhan’ *R. Cowley, F. Caprotti, M. Ferretti, C. Zhong - Inside smart cities pp.45-64, 2018*
- ‘An Exploratory Analysis of Ethnic Groups in the City of Milan through Mobile Phone Data’ *G.Barlacchi, M.Ferretti, B.Lepri and F.Manfredini - NetMob, 2017*
- ‘A Multi-Scale Approach to Data-Driven Mass Migration Analysis’ *M. Ahmed, G. Barlacchi, S. Braghin, F. Calabrese, M. Ferretti, V. Lonij, R. Nair, R. Novack, J. Paraszczak, A. S. Toor - SoGood@ ECML-PKDD, 2017*

HONOURS AND AWARDS

- King’s Award for Excellence in Innovation and Impact, KCL, 2017;
- DataSpark Mobility Challenge (2nd place), International Conference on Information and Knowledge Management (CIKM), Singapore 2017;
- The “Professor Sir Richard Trainor” PhD scholarship funding, KCL, 2015;
- Outstanding Thesis Award “Stefano Pieracci”, XVIII^o Edition, AISRe - ERSa Italy chapter, 2013;
- “Sistemi Informativi Territoriali e Governo del Territorio” postgraduate course scholarship, Politecnico di Milano, Italy, 2011.